IPCON® PC FR HS

Flame Retardant High Speed Polycarbonate Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	340 ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 110 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.22 g/cm³
Shrinkage Rate*	Specially-defined	0.39% - 0.50%
Surface Hardness	Shore hardness	84D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	0.28%
Melt Index	255 °C, 2.16 kg	19.4 g/10 min
Melting Temperature	DSC, 10 °C/min	208 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	97 °C
Heat Deflection Temperature	ISO 75, 0.45 MPa	92 °C

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	56 MPa
Tensile Strength Z	ISO 527, GB/T 1040	42 MPa
Young's Modulus XY	ISO 527, GB/T 1040	2137 MPa
Young's Modulus Z	ISO 527, GB/T 1040	1827 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	12.3%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	4.7%
Flexural Strength XY	ISO 178, GB/T 9341	101 MPa
Flexural Strength Z	ISO 178, GB/T 9341	75 MPa
Flexural Modulus XY	ISO 178, GB/T 9341	2521 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	2754 MPa
Impact Strength XY	ISO 179, GB/T 1043	62.6 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	11.2 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 255 °C, bed temperature: 95 °C, printing speed: XY—240 mm/s and Z—100 mm/s) and IPCON PC FR HS (Navy Blue) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill

pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. The world's first fluorine-free and halogen-free UL94 V-0 @0.8 mm ultra-thin flame-retardant and translucent PC filament; national trend aesthetics, transparency and color depth are adjustable with thickness; high strength, impact resistance, low warpage; compatible with AMS or CFS, and can be printed at ultra-high speeds.
2. Can be used to print consumer electronics products that require top-level flame retardancy, being fluorine-free and halogen-free, such as smart lamps, mobile power banks, network patch panels and computer hard drive cage casings, as well as high-strength and heat-resistant components such as outdoor stove tripods.

• Recommended Printing Settings

Drying Settings Before Printing	65 - 75 °C, 8 - 12 h (Blast drying oven)
Printing Temperature and Humidity	≤ 60 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PC
Compatible Printer Type	Enclosed-frame (recommended), open-frame
AMS or CFS	Compatible
Compatible Nozzle Material	Any common material, such as brass, stainless steel, hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.2 / 0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick
Nozzle Temperature	240 - 270 °C
Bed Temperature	85 - 100 °C
Fan Speed	0% - 60%
Max Print Speed	350 mm/s

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